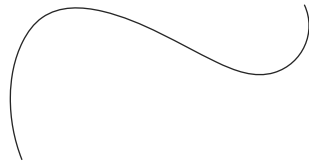


Curves and Surfaces

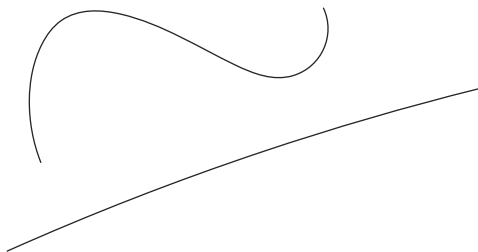
First we will be studying the curvature of curves

Curves and Surfaces



For instance, I have drawn a curve here

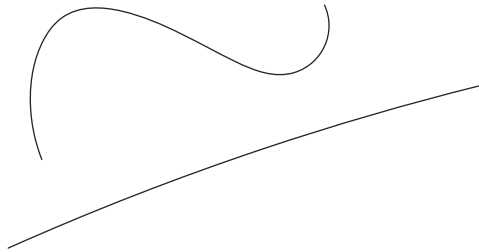
Curves and Surfaces



Even a straight line will be called a curve, but with “curvature” 0

Curves and Surfaces

What is a curve?

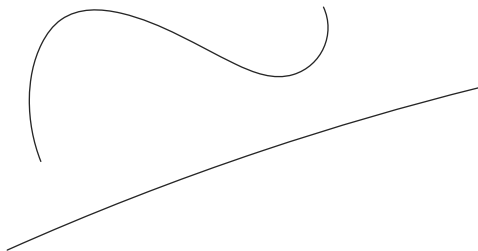


I asked for suggestions to define such objects.

Curves and Surfaces

What is a curve?

1. Path?

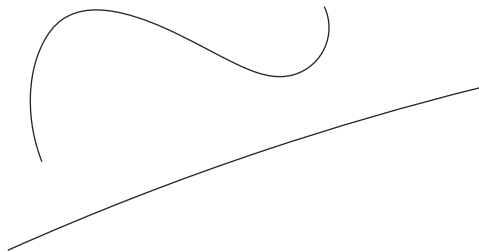


One suggestion was that a curve is a path.

Curves and Surfaces

What is a curve?

1. Path?

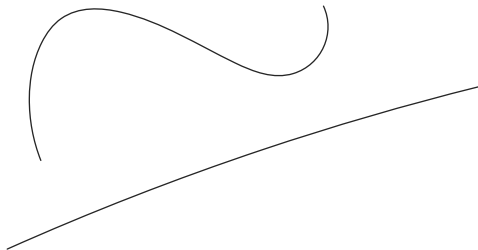


A good start. How do we now define a path mathematically?

Curves and Surfaces

What is a curve?

1. Path?
2. One dimensional space

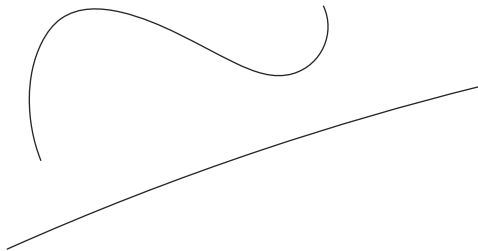


Another suggestion was that it is one dimensional. How do we define dimension?

Curves and Surfaces

What is a curve?

1. Path?
2. One dimensional space
3. It is a set of points that satisfies some equation

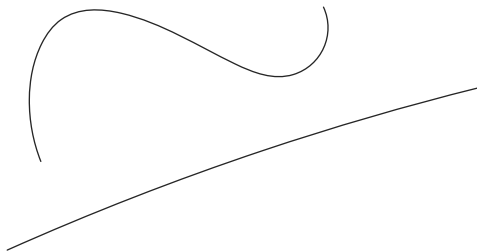


Another was that it is a set of points that satisfies some equation.

Curves and Surfaces

What is a curve?

1. Path?
2. One dimensional space
 - a) It has only length
 - b) It has only one coordinate
3. It is a set of points that satisfies some equation

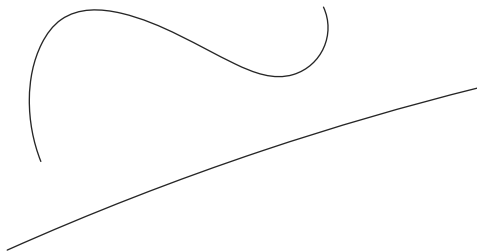


One of you suggested that one dimensional could mean that it has only length or one coordinate

Curves and Surfaces

What is a curve?

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3. It is a set of points that satisfies some equation

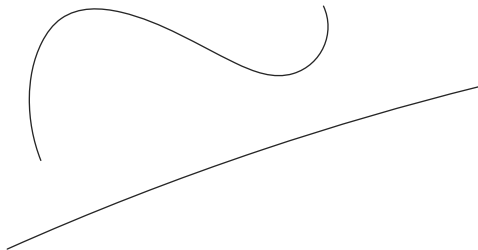


Again, a good start but we need to translate these to precise mathematical language

Curves and Surfaces

What is a curve?

1. Path?
2. One dimensional space
 - a) It has only length
 - b) It has only one coordinate
3. It is a set of points that satisfies some equation
4. Can have only one direction of motion

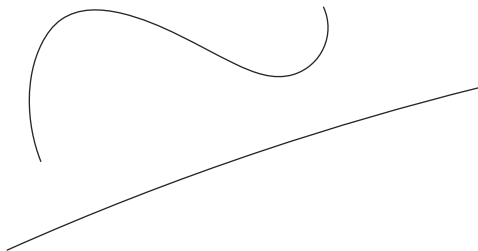


Yet another suggestion was to think of a path as being traced by something moving!

Curves and Surfaces

What is a curve?

1. Path?
2. One dimensional space
 - a) It has only length
 - b) It has only one coordinate
3. It is a set of points that satisfies some equation
4. Can have only one direction of motion
5. Looks like $\mathbb{R}$ from inside even if it is curved from the outside

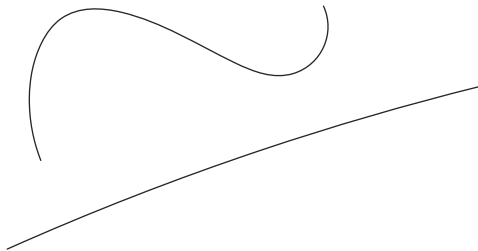


Another suggestion emphasized that even though it was curved in how it was situated in space

Curves and Surfaces

What is a curve?

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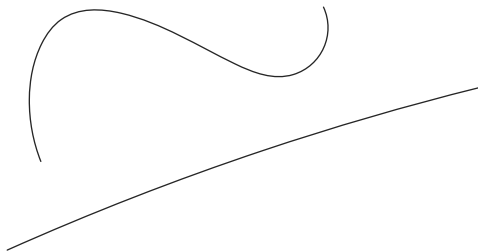
it is not itself curved

Curves and Surfaces

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Examples



Let us look at some examples

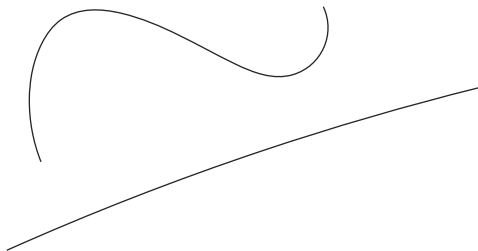
Curves and Surfaces

What is a curve?

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 - a) It has only length
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3. It is a set of points that satisfies some equation
4. Can have only one direction of motion
5. Looks like $\mathbb{R}$ from inside even if it is curved from the outside

Examples

How would you represent a line?



Start with the simplest: a line.

Curves and Surfaces

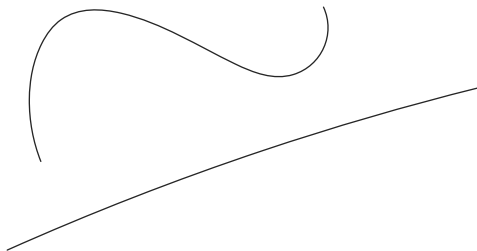
What is a curve?

1. Path?
2. One dimensional space
 - a) It has only length
 - b) It has only one coordinate
3. It is a set of points that satisfies some equation
4. Can have only one direction of motion
5. Looks like $\mathbb{R}$ from inside even if it is curved from the outside

Examples

How would you represent a line?

$$y = x + 3$$



One way to represent a line would be to use the highlighted definition

Curves and Surfaces

What is a curve?

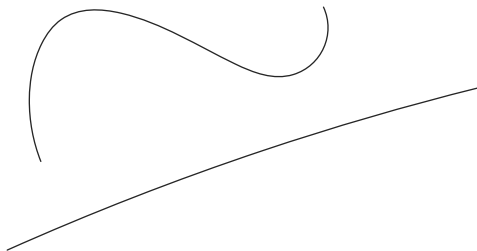
1. Path?
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 - a) It has only length
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4. Can have only one direction of motion
5. Looks like $\mathbb{R}$ from inside even if it is curved from the outside

Examples

How would you represent a line?

$$y = x + 3$$

(a, b)



So what does that tell you about a point on the line?

Curves and Surfaces

What is a curve?

1. Path?
2. One dimensional space
 - a) It has only length
 - b) It has only one coordinate**
3. It is a set of points that satisfies some equation
4. Can have only one direction of motion
5. Looks like $\mathbb{R}$ from inside even if it is curved from the outside

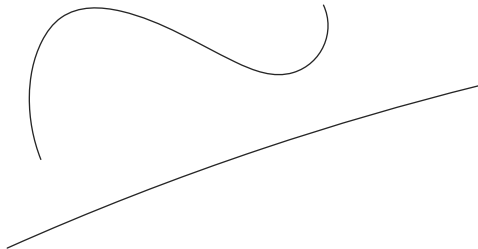
Examples

How would you represent a line?

$$y = x + 3$$

$$(a, b)$$

$$(a, a + 3)$$



One of you observed that although there are two variables, only one is independent.

Curves and Surfaces

What is a curve?

1. Path?
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 - a) It has only length
 - b) It has only one coordinate**
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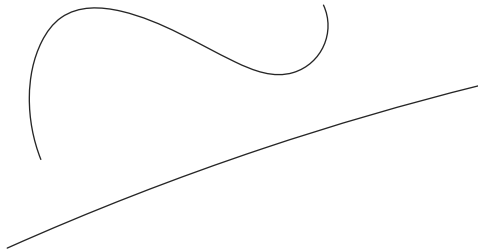
Examples

How would you represent a line?

$$y = x + 3$$

$$(a, b)$$

$$(a, a + 3)$$



which does explain the meaning of “one coordinate” in the highlighted definition.

Curves and Surfaces

What is a curve?

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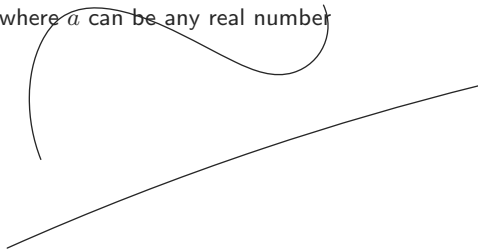
Examples

How would you represent a line?

$$y = x + 3$$

(a, b)

$(a, a + 3)$ where a can be any real number



So this line is the set of points of the highlighted form

Curves and Surfaces

What is a curve?

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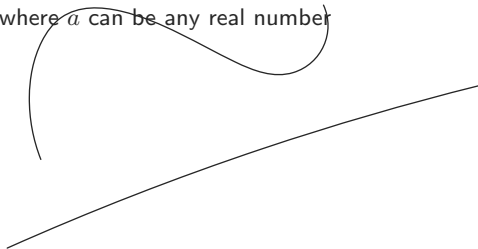
Examples

How would you represent a line?

$$y = x + 3$$

(a, b)

$(a, a + 3)$ where a can be any real number



where 'a' can be any real number

Curves and Surfaces

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$$(f(t), g(t))$$

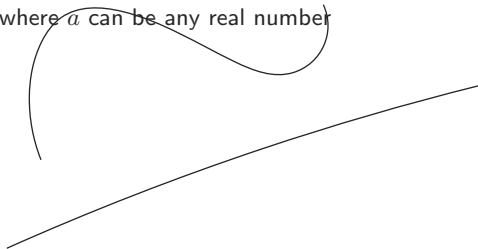
Examples

How would you represent a line?

$$y = x + 3$$

$$(a, b)$$

$(a, a + 3)$ where a can be any real number



So we can say that a curve is a set of points in $\mathbb{R}^2$

Curves and Surfaces

What is a curve?

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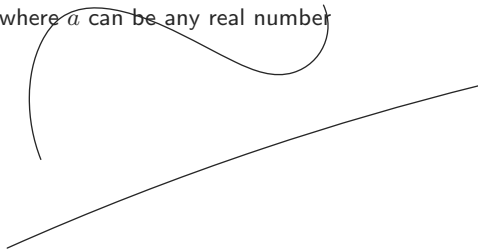
Examples

How would you represent a line?

$$y = x + 3$$

$$(a, b)$$

$(a, a + 3)$ where a can be any real number



whose coordinates are functions of a single parameter t

Curves and Surfaces

What is a curve?

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$(f(t), g(t))$ for each $t \in \mathbb{R}$

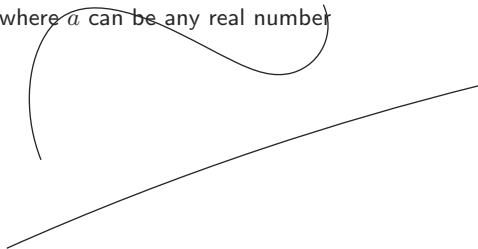
Examples

How would you represent a line?

$$y = x + 3$$

$$(a, b)$$

$(a, a + 3)$ where a can be any real number



and the parameter can vary over all real numbers

Curves and Surfaces

What is a curve?

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$(f(t), g(t))$ for each $t \in \mathbb{R}$

A parametrization is a function $\theta : (a, b) \rightarrow \mathbb{R}^2$

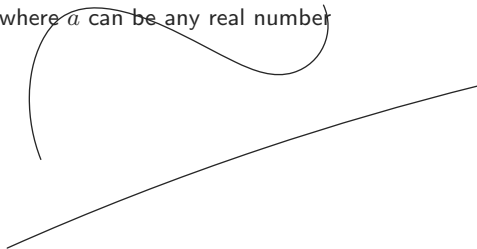
Examples

How would you represent a line?

$$y = x + 3$$

$$(a, b)$$

$(a, a + 3)$ where a can be any real number



But why restrict our attention to curves that are indefinite

Curves and Surfaces

What is a curve?

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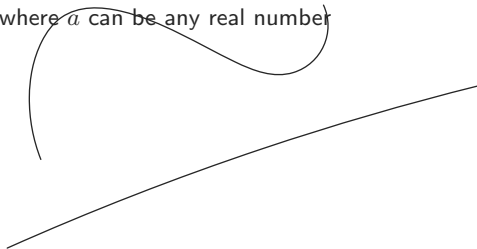
Examples

How would you represent a line?

$$y = x + 3$$

$$(a, b)$$

$(a, a + 3)$ where a can be any real number



So the parameter t can be from any interval. That interval may be the whole $\mathbb{R}$

Curves and Surfaces

What is a curve?

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Examples:

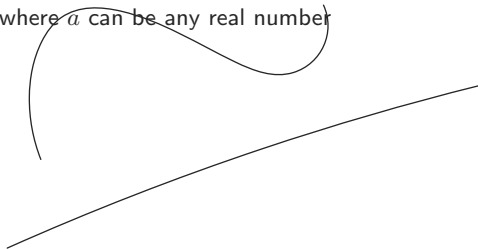
Examples

How would you represent a line?

$$y = x + 3$$

$$(a, b)$$

$(a, a + 3)$ where a can be any real number



Let us now consider more examples

Curves and Surfaces

What is a curve?

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4. Can have only one direction of motion
5. Looks like $\mathbb{R}$ from inside even if it is curved from the outside

$(f(t), g(t))$ for each $t \in \mathbb{R}$

A parametrization is a function $\theta : (a, b) \rightarrow \mathbb{R}^2$

Examples:

1. $t \rightarrow (t + 2, 4t)$

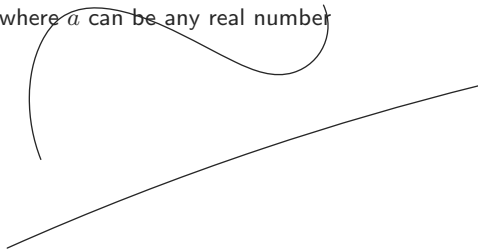
Examples

How would you represent a line?

$$y = x + 3$$

$$(a, b)$$

$(a, a + 3)$ where a can be any real number



Here is another line

Curves and Surfaces

What is a curve?

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5. Looks like $\mathbb{R}$ from inside even if it is curved from the outside

$(f(t), g(t))$ for each $t \in \mathbb{R}$

A parametrization is a function $\theta : (a, b) \rightarrow \mathbb{R}^2$

Examples:

1. $t \rightarrow (t + 2, 4t)$
2. $t \rightarrow (\cos(t), \sin(t))$

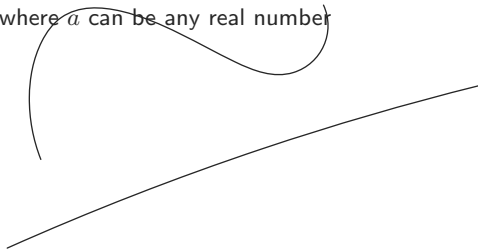
Examples

How would you represent a line?

$$y = x + 3$$

$$(a, b)$$

$(a, a + 3)$ where a can be any real number



This is how one could parametrize a circle of radius 1

Curves and Surfaces

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Examples:

1. $t \rightarrow (t + 2, 4t)$
2. $t \rightarrow (\cos(t), \sin(t))$
3. $t \rightarrow (t, at^2)$

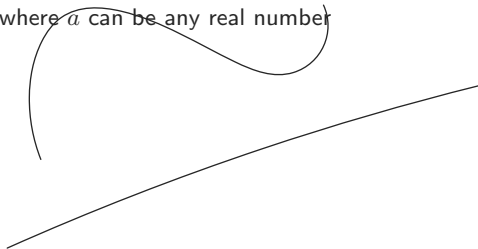
Examples

How would you represent a line?

$$y = x + 3$$

$$(a, b)$$

$(a, a + 3)$ where a can be any real number



And one of you correctly suggested this to parametrize a parabola

Curves and Surfaces

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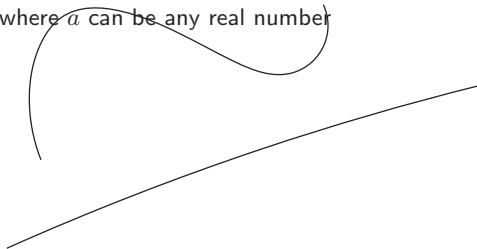
Examples

How would you represent a line?

$$y = x + 3$$

$$(a, b)$$

$(a, a + 3)$ where a can be any real number



But do you always get something of the same dimension?

Curves and Surfaces

What is a curve?

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$(f(t), g(t))$ for each $t \in \mathbb{R}$

A parametrization is a function $\theta : (a, b) \rightarrow \mathbb{R}^2$

Examples:

1. $t \rightarrow (t + 2, 4t)$
2. $t \rightarrow (\cos(t), \sin(t))$
3. $t \rightarrow (t, at^2)$

We can get a point if f and g are constants

As you correctly pointed out, if f and g are constants the dimension will be 0

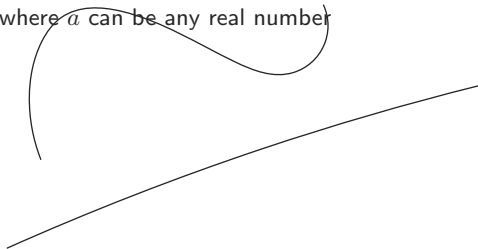
Examples

How would you represent a line?

$$y = x + 3$$

$$(a, b)$$

$(a, a + 3)$ where a can be any real number



Curves and Surfaces

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Examples:

1. $t \rightarrow (t + 2, 4t)$
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3. $t \rightarrow (t, at^2)$

We can get a point if f and g are constants

We demand that the derivatives

non-zero

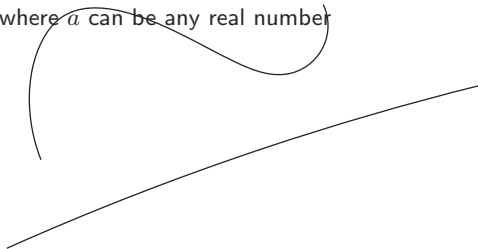
Examples

How would you represent a line?

$$y = x + 3$$

$$(a, b)$$

$(a, a + 3)$ where a can be any real number



So we demand that the derivatives be non-zero

Curves and Surfaces

What is a curve?

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$(f(t), g(t))$ for each $t \in \mathbb{R}$

A parametrization is a function $\theta : (a, b) \rightarrow \mathbb{R}^2$

Examples:

1. $t \rightarrow (t + 2, 4t)$
2. $t \rightarrow (\cos(t), \sin(t))$ $t \in (0, 2\pi)$
3. $t \rightarrow (t, at^2)$

We can get a point if f and g are constants

We demand
non-zero

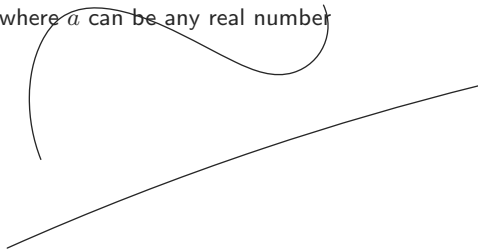
Examples

How would you represent a line?

$$y = x + 3$$

$$(a, b)$$

$(a, a + 3)$ where a can be any real number



Notice then that theta is pushing an interval, which is certainly one-dimensional, into $\mathbb{R}^2$

Curves and Surfaces

What is a curve?

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$(f(t), g(t))$ for each $t \in \mathbb{R}$

A parametrization is a function $\theta : (a, b) \rightarrow \mathbb{R}^2$ where f and g have non-zero derivatives

Examples:

1. $t \rightarrow (t + 2, 4t)$
2. $t \rightarrow (\cos(t), \sin(t))$ $t \in (0, 2\pi)$
3. $t \rightarrow (t, at^2)$

We can get a point if f and g are constants

We demand that the de

non-zero

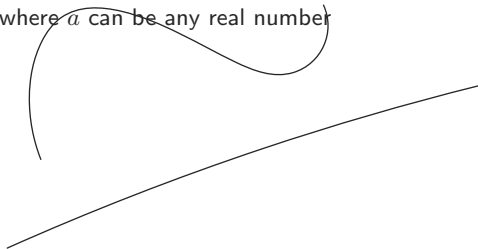
Examples

How would you represent a line?

$$y = x + 3$$

$$(a, b)$$

$(a, a + 3)$ where a can be any real number



So we add the condition that f and g have non-zero derivatives.

Curves and Surfaces

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$(f(t), g(t))$ for each $t \in \mathbb{R}$

A parametrization is a function $\theta : (a, b) \rightarrow \mathbb{R}^2$ where f and g have non-zero derivatives

Examples:

1. $t \rightarrow (t + 2, 4t)$
2. $t \rightarrow (\cos(t), \sin(t))$ $t \in (0, 2\pi)$
3. $t \rightarrow (t, at^2)$

We can get a point if f and g are constants

We demand that the derivative is non-zero

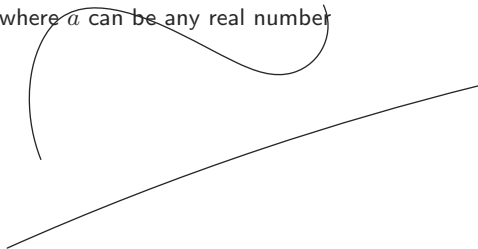
Examples

How would you represent a line?

$$y = x + 3$$

$$(a, b)$$

$(a, a + 3)$ where a can be any real number



Of course we implicitly mean that f and g are differentiable

Curves and Surfaces

What is a curve?

1. Path?
2. One dimensional space
 - a) It has only length
 - b) It has only one coordinate
3. It is a set of points that satisfies some equation
4. Can have only one direction of motion
5. Looks like $\mathbb{R}$ from inside even if it is curved from the outside

$(f(t), g(t))$ for each $t \in \mathbb{R}$

A parametrization is a function $\theta : (a, b) \rightarrow \mathbb{R}^2$ where f and g have non-zero derivatives

Examples:

1. $t \rightarrow (t + 2, 4t)$
2. $t \rightarrow (\cos(t), \sin(t))$ $t \in (0, 2\pi)$
3. $t \rightarrow (t, at^2)$

We can get a point if f and g are constants

We demand that f and g are

non-zero

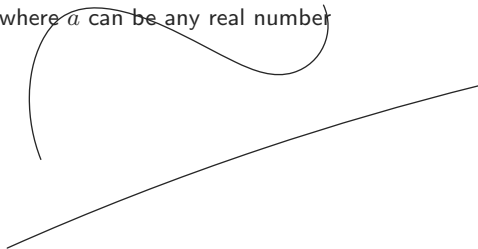
Examples

How would you represent a line?

$$y = x + 3$$

$$(a, b)$$

$(a, a + 3)$ where a can be any real number



It is easy to see that this ensures that the dimension of the image is never 0

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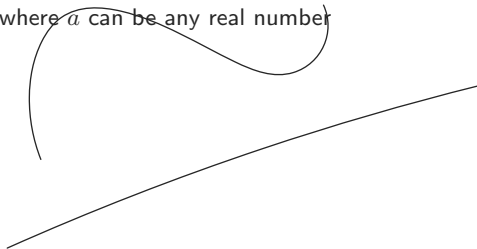
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It also ensures that the dimension can never be larger

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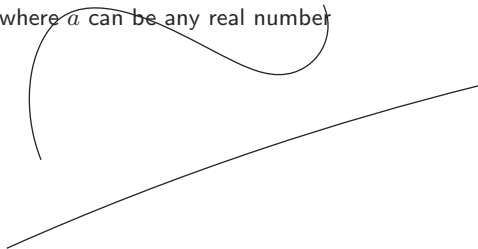
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Though that takes a bit more care

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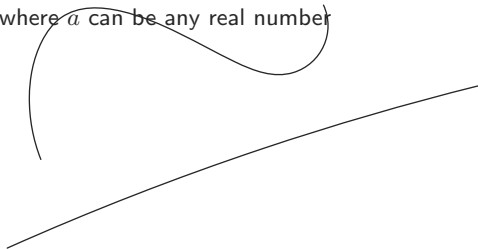
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Notice that the definition in blue also serves as a good definition of a path

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We demand that the derivatives of θ are non-zero from the image of a a to the image of b

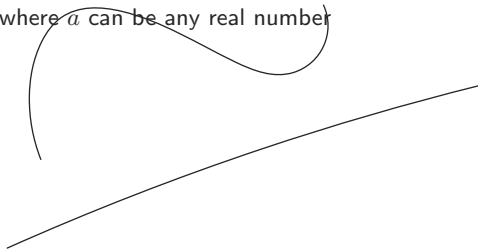
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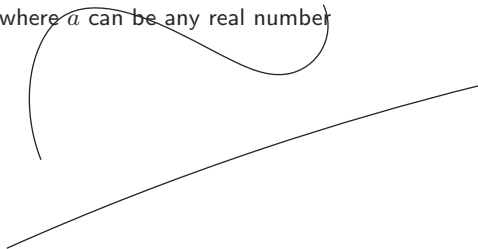
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It helps here to think of t as a time parameter

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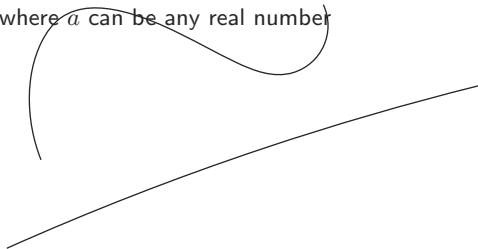
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And the parametrization as a function that describes how you drew that curve

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We demand that the derivatives of f and g in some interval of time non-zero

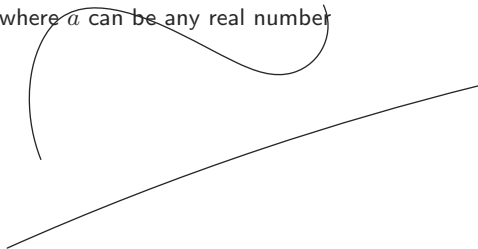
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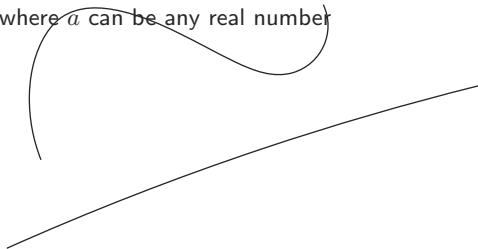
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But we will see that all your other suggestions are also taken care of!

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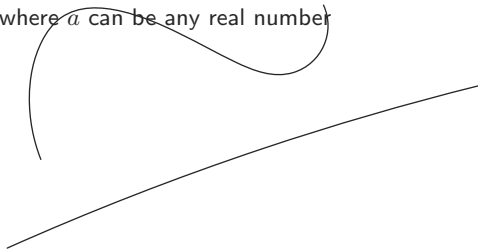
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We know that the interval (a, b) is certainly one dimensional

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We demand that the derivatives be non-zero because it is a part of $\mathbb{R}$ which is just one axis

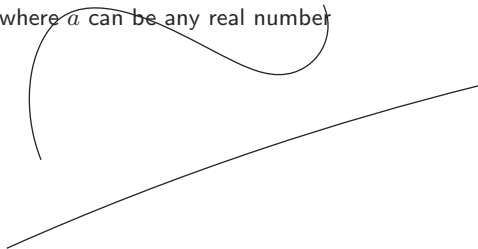
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We demonstrate that we have chosen the parametrization so that it never collapses the dimension nor enlarges it non-zero

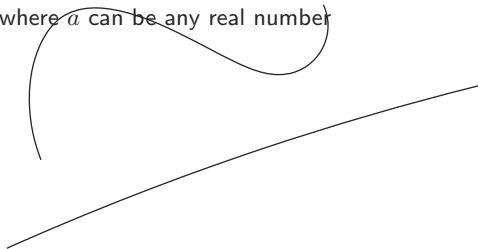
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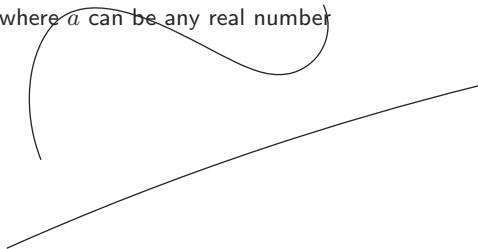
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In fact now we can see what it means to have only length

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The interval certainly has only length because it lives inside just one axis of the coordinate system

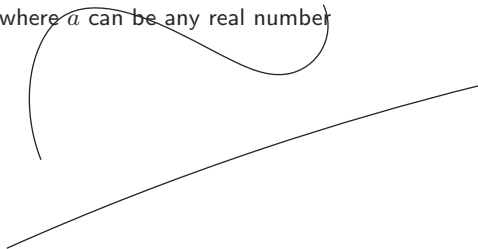
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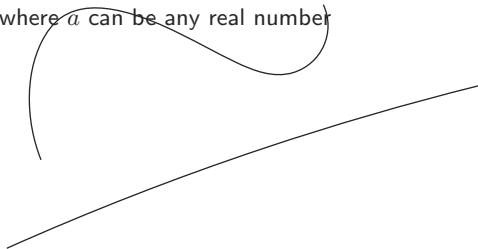
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That also gives meaning to the second point

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We which is that it has only one coordinate. So that "coordinate" referred to the domain not the coodomain.
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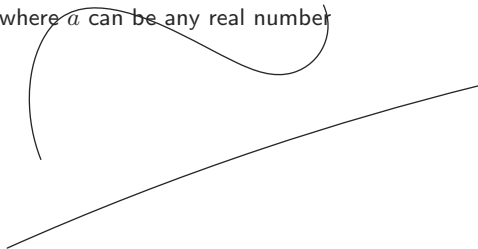
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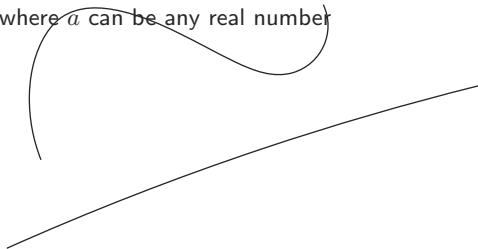
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You need two coordinates, $f(t)$, $g(t)$ to push it in $\mathbb{R}^2$ but only one t .

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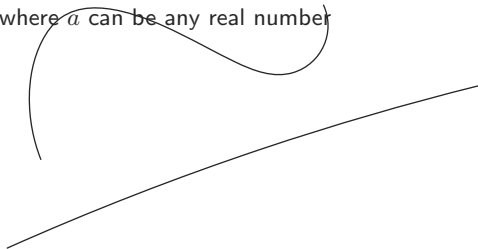
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This one is a bit subtle. If it satisfies one equation does it mean we can parametrize it?

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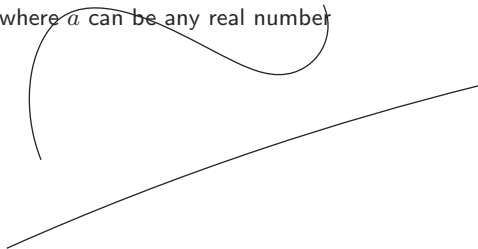
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Yes! But perhaps only in the vicinity that we are studying

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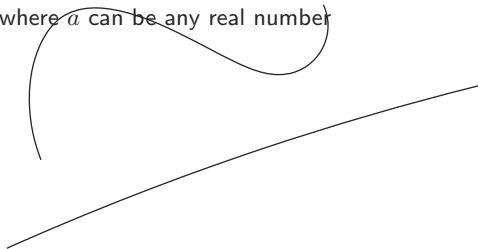
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More on that later

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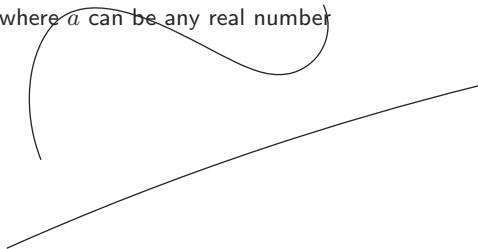
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This suggested definition is captured very easily!

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$(f(t), g(t))$ for each $t \in \mathbb{R}$

A parametrization is a function $\theta : (a, b) \rightarrow \mathbb{R}^2$ where f and g have non-zero derivatives

Examples:

1. $t \rightarrow (t + 2, 4t)$
2. $t \rightarrow (\cos(t), \sin(t))$ $t \in (0, 2\pi)$
3. $t \rightarrow (t, at^2)$

We can get a point if f and g are constants

Why? because the parametrization may be thought of as describing the motion of a point in one direction of time non-zero

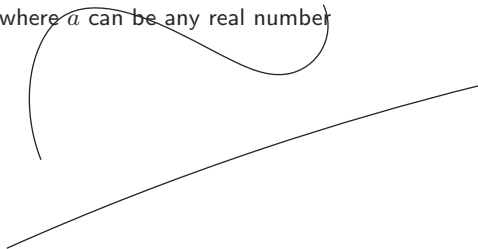
Examples

How would you represent a line?

$$y = x + 3$$

$$(a, b)$$

$(a, a + 3)$ where a can be any real number



Curves and Surfaces

What is a curve?

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We demand that the
non-zero

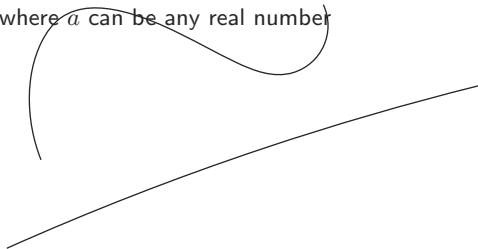
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As for this suggestion, we can think of (a, b) as what the curve really is

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We do not get a curve if f and g are non-zero

And the parametrization pushes it into $\mathbb{R}^2$ by curving it so that to other points in $\mathbb{R}^2$ it looks curved

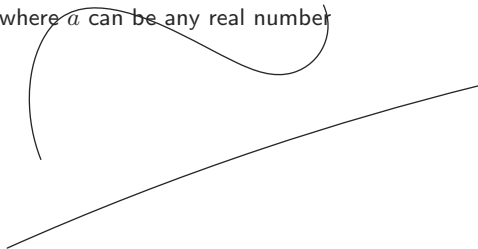
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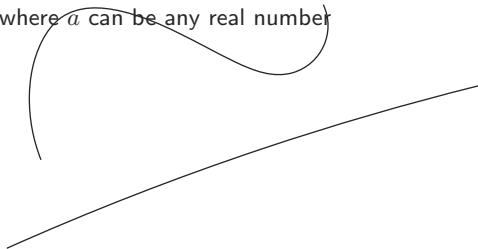
Examples

How would you represent a line?

$$y = x + 3$$

$$(a, b)$$

$(a, a + 3)$ where a can be any real number



Our definition gives us another advantage. Consider the highlighted definition

Curves and Surfaces

$$x^2 + y^2 + z^2 = 1$$

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We demand that the derivatives

non-zero

If I allow 3 variables, it no longer defines a curve.

Curves and Surfaces

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In this example, the points satisfying the given equation is a sphere of radius 1

Curves and Surfaces

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We can get a point if f and g are constants

We demand that the derivative is non-zero

because this time there are 3 variables, so it lives in $\mathbb{R}^3$

Curves and Surfaces

$$x^2 + y^2 + z^2 = 1 \text{ and } z = x + y$$

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To get a curve, we would need another equation.

Curves and Surfaces

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We can get a point if f and g are constants

We demand that t

non-zero

For example if we demand that it must also satisfy the equation of a plane

Curves and Surfaces

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We can get a point if f and g are constants

We demand that the derivatives of f and g are non-zero then it lies on both a sphere and a plane

Curves and Surfaces

$$x^2 + y^2 + z^2 = 1 \text{ and } z = x + y$$

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We can get a point if f and g are constants

We demand that the derivatives of f and g are non-zero and a plane slices a sphere in a circle

Curves and Surfaces

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We can get a point if f and g are constants

We demand that the derivatives of f and g are non-zero so this gives a circle in 3 space

Curves and Surfaces

$$x^2 + y^2 + z^2 = 1 \text{ and } z = x + y$$

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We can get a point if f and g are constants

We demand that the derivative

non-zero

This way of looking at curves is very fruitful in another area

Curves and Surfaces

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We can get a point if f and g are constants

We demand that the derivative is non-zero

But for us, our parametrization makes things much easier

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We can get a point if f and g are constants

We demand that the derivative is non-zero

$$x^2 + y^2 + z^2 = 1 \text{ and } z = x + y$$

Even in $\mathbb{R}^3$:

A parametrization is a function $\theta : (a, b) \rightarrow \mathbb{R}^3$, say $(f(t), g(t), h(t))$ where f , g , and h have non-zero derivatives

Our definition easily carries over to curves in 3 space too!

Curves and Surfaces

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All one has to do is to replace the $\mathbb{R}^2$ by $\mathbb{R}^3$

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We can get a point if f and g are constants

We can't have a curve if f and g are constants. Having pinned down our definition of a curve, let us try to figure out what it means for it to be curved

$$x^2 + y^2 + z^2 = 1 \text{ and } z = x + y$$

Even in $\mathbb{R}^3$:

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How do we know if a curve is curved?

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We demand that
non-zero

$$x^2 + y^2 + z^2 = 1 \text{ and } z = x + y$$

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How do we know if a curve is curved?

Check if the direction is changing

One of your correctly suggested that we should check if the direction is changing

Curves and Surfaces

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How do we know if a curve is curved?

Check if the direction is changing

But how do we ensure that?

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How do we know if a curve is curved?

Check if the direction is changing

Keep the speed constant and measure the acceleration

One way is to keep the speed constant.

Curves and Surfaces

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How do we know if a curve is curved?

Check if the direction is changing

Keep the speed constant and measure the acceleration

Remember the difference between speed and velocity

Curves and Surfaces

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Check if the direction is changing

Keep the speed constant and measure the acceleration

Velocity also involves direction.

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How do we know if a curve is curved?

Check if the direction is changing

Keep the speed constant and measure the acceleration

If there is acceleration, which means the velocity is changing

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How do we know if a curve is curved?

Check if the direction is changing

Keep the speed constant and measure the acceleration

But the speed is not, it means that the direction is changing!

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We can get a point if f and g are constants

We If our definition is correct, then if we consider a parametrization with constant speed and 0 acceleration non-zero

$$x^2 + y^2 + z^2 = 1 \text{ and } z = x + y$$

Even in $\mathbb{R}^3$:

A parametrization is a function $\theta : (a, b) \rightarrow \mathbb{R}^3$, say $(f(t), g(t), h(t))$ where f , g , and h have non-zero derivatives

How do we know if a curve is curved?

Check if the direction is changing

Keep the speed constant and measure the acceleration

Exercise: Show that if the acceleration of a "constant speed parametrization" is 0, then the curve is a line

Curves and Surfaces

What is a curve?

1. Path?
2. One dimensional space
 - a) It has only length
 - b) It has only one coordinate
3. It is a set of points that satisfies some equation
4. Can have only one direction of motion
5. Looks like $\mathbb{R}$ from inside even if it is curved from the outside

$(f(t), g(t))$ for each $t \in \mathbb{R}$

A parametrization is a function $\theta : (a, b) \rightarrow \mathbb{R}^2$ where f and g have non-zero derivatives

Examples:

1. $t \rightarrow (t + 2, 4t)$
2. $t \rightarrow (\cos(t), \sin(t))$ $t \in (0, 2\pi)$
3. $t \rightarrow (t, at^2)$

We can get a point if f and g are constants

We demand that f and g have non-zero derivatives, then the curve it traces must be a straight line. This is a simple exercise for you!

$$x^2 + y^2 + z^2 = 1 \text{ and } z = x + y$$

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What is the speed of a parametrization θ ?

To formalize our definition we have to carefully define each term. Let us start with speed.

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We can get a point if f and g are constants

We can get a non-zero velocity if f and g are non-constant. Since the velocity is just a derivative, we simply take the magnitude of the derivative to get the speed

$$x^2 + y^2 + z^2 = 1 \text{ and } z = x + y$$

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What is the speed of a parametrization θ ?

We take the magnitude of the derivative

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We demand that the derivatives of f and g are non-zero

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We take the magnitude of the derivative

$$\sqrt{f'^2(t) + g'^2(t)}$$

So this is the formula

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What is the speed of a parametrization θ ?

We take the magnitude of the derivative

$$\|\theta'(t)\| = \sqrt{f'^2(t) + g'^2(t)}$$

And the magnitude is denoted by wrapping it in double bars

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$$\|\theta'(t)\| = \sqrt{f'^2(t) + g'^2(t)}$$

What is the acceleration of θ

Let us now try to define the acceleration

Curves and Surfaces

We take a constant speed parametrization and consider the double derivative to obtain the curvature

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What is the acceleration of θ

$$\theta''(t) = (f''(t), g''(t))$$

So this is our definition of curvature We just need to differentiate the velocity again, giving us the double derivative

Curves and Surfaces

We take a constant speed parametrization and consider the double derivative to obtain the curvature

Consider a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$ such that $\|\theta'(t)\|$ is constant over t , then the curvature is $\|\theta''(t)\|$

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What is the acceleration of θ

$$\theta''(t) = (f''(t), g''(t))$$

Which can be written more mathematically as this

Curves and Surfaces

We take a constant speed parametrization and consider the double derivative to obtain the curvature

Consider a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$ such that $\|\theta'(t)\|$ is constant over t , then the curvature is $\|\theta''(t)\|$

Example: Consider a parabola

$$x^2 + y^2 + z^2 = 1 \text{ and } z = x + y$$

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What is the acceleration of θ

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But do you sense a problem? Consider a parabola.

Curves and Surfaces

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Consider a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$ such that $\|\theta'(t)\|$ is constant over t , then the curvature is $\|\theta''(t)\|$

Example: Consider a parabola $t \rightarrow (t, t^2)$

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which is given by this parametrization

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Example: Consider a parabola $t \rightarrow (t, t^2)$. It is not of constant speed.

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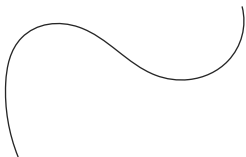
This parametrization is not of constant speed!

Curves and Surfaces

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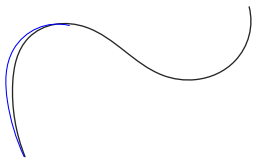
In general, I may not have drawn a curve with a constant speed parametrization

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We take a constant speed parametrization and consider the double derivative to obtain the curvature

Consider a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$ such that $\|\theta'(t)\|$ is constant over t , then the curvature is $\|\theta''(t)\|$

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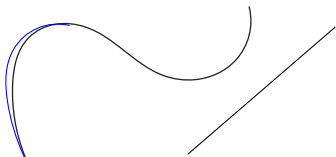
In a constant speed one, let us say constant unit speed, the length of the path traversed would match with the time

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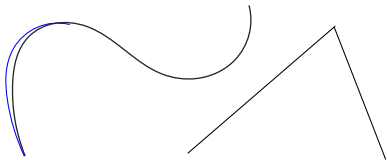
One of you detected another problem

Curves and Surfaces

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Example: Consider a parabola $t \rightarrow (t, t^2)$. It is not of constant speed.



If the curve is bent like this, the double derivative, and therefore the acceleration is not defined.

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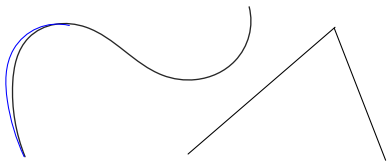
Curves and Surfaces

We take a constant speed parametrization and consider the double derivative to obtain the curvature

Consider a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$ such that $\|\theta'(t)\|$ is constant over t , then the curvature is $\|\theta''(t)\|$

Example: Consider a parabola $t \rightarrow (t, t^2)$. It is not of constant speed.

Does every curve have a constant speed parametrization?



$$x^2 + y^2 + z^2 = 1 \text{ and } z = x + y$$

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To resolve the first problem, we need to ask if we can always find some constant speed parametrization

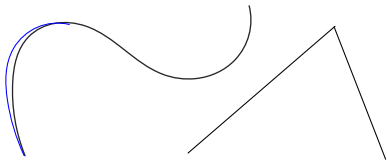
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We take a constant speed parametrization and consider the double derivative to obtain the curvature

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Even if we did not begin with one!

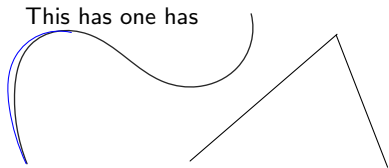
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Does every curve have a constant speed parametrization?



This curve has a constant speed parametrization

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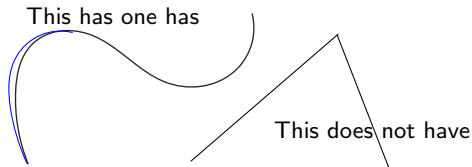
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Example: Consider a parabola $t \rightarrow (t, t^2)$. It is not of constant speed.

Does every curve have a constant speed parametrization?



Whereas this curve cannot have a constant speed parametrization because of its corner

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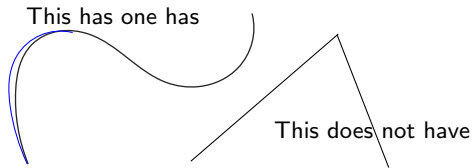
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Example: Consider a parabola $t \rightarrow (t, t^2)$. It is not of constant speed.

Does every curve have a constant speed parametrization?



Even in $\mathbb{R}^3$:

A parametrization is a function $\theta : (a, b) \rightarrow \mathbb{R}^3$, say $(f(t), g(t), h(t))$ where f , g , and h have non-zero derivatives and all the derivatives exist

which is that at least the derivatives upto the second derivative must exist So we need to add something to this def

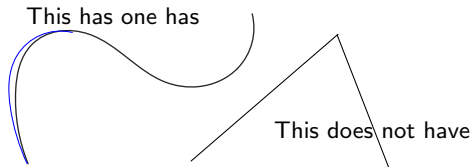
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but we will ensure that all the derivatives exist, to be safer

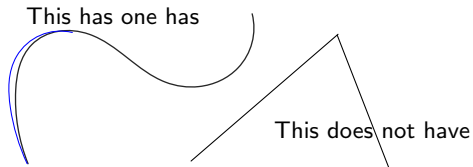
Curves and Surfaces

We take a constant speed parametrization and consider the double derivative to obtain the curvature

Consider a parametrization $\theta : (a, b) \rightarrow \mathbb{R}^2$ such that $\|\theta'(t)\|$ is constant over t , then the curvature is $\|\theta''(t)\|$

Example: Consider a parabola $t \rightarrow (t, t^2)$. It is not of constant speed.

Does every curve have a constant speed parametrization?



Even in $\mathbb{R}^3$:

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Such functions or parametrizations for which all derivatives exist, are called smooth

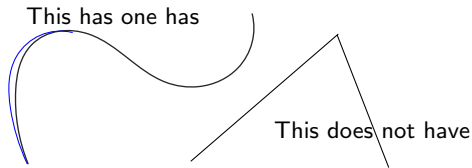
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A digression: Consider these two concepts:

Let us now consider a very very important concept in geometry

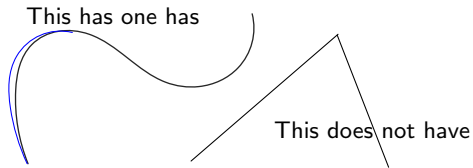
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A digression: Consider these two concepts:

1. The distance between two points on plane

You can use coordinate geometry to calculate the distance between two points

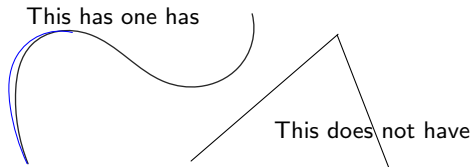
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1. The distance between two points on plane

So you do impose coordinates

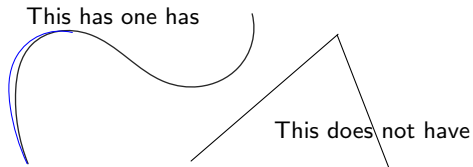
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But given your coordinates you can also define the sum of the x coordinates!

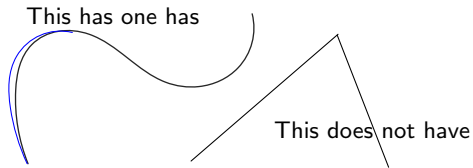
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Which of these is a “geometric” concept and which of these are “coordinate dependent”

Yet only one of these concepts is truly geometric in a way that we can now make precise

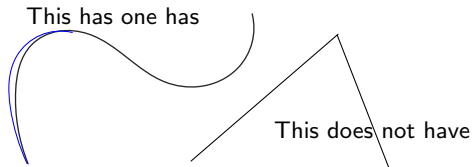
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Does every curve have a constant speed parametrization?



Only the first one remains the same when we change coordinates!!

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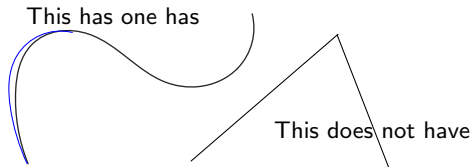
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Try to check that the formula for distance would give you the same answer

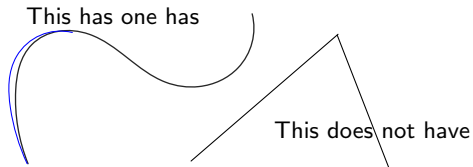
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Does every curve have a constant speed parametrization?



no matter where you decide to place the origin of the coordinates.

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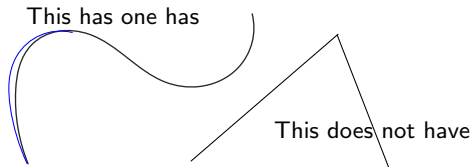
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That is why only the highlighted concept is a geometric one

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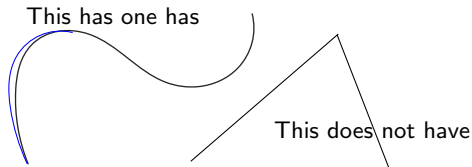
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The other is specifically about the coordinate system we used, and not the underlying geometry

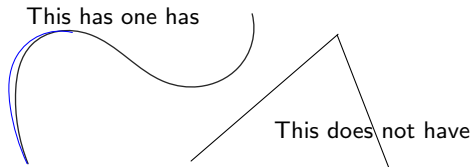
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If we are introducing a geometric concept it must be invariant under a coordinate change

Any time we define a geometric concept we have to ensure it is invariant under coordinate change

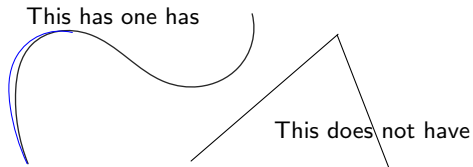
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But it is not just the coordinates system chosen to represent points of $\mathbb{R}^3$ that is a problem in our definition

$$t \rightarrow (\cos(t), \sin(t))$$

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Consider the circle which can be parametrized like this Even the parametrization is arbitrarily chosen

$t \rightarrow (\cos(t), \sin(t))$ where $t \in (0, 2\pi)$

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in which case we would need to use the interval 0 to 2π

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$t \rightarrow (\cos(2t), \sin(2t))$

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But this parametrization also traces a circle, just at twice the speed!

$t \rightarrow (\cos(t), \sin(t))$ where $t \in (0, 2\pi)$

$t \rightarrow (\cos(2t), \sin(2t))$ where $t \in (0, \pi)$

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and therefore needs only half the interval to trace the circle

$t \rightarrow (\cos(t), \sin(t))$ where $t \in (0, 2\pi)$

$t \rightarrow (\cos(2t), \sin(2t))$ where $t \in (0, \pi)$

Both are valid parametrizations of the same curve

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So there are many parametrizations possible for a given curve

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How are two parametrizations of the same curve related?

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You need to think about how two parametrizations of the curves are related.

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Once we do that, we have to ensure that our definition of acceleration

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does not depend on the particular parametrization we chose

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All this will be discussed the next time. In the mean while try to think about it yourself.